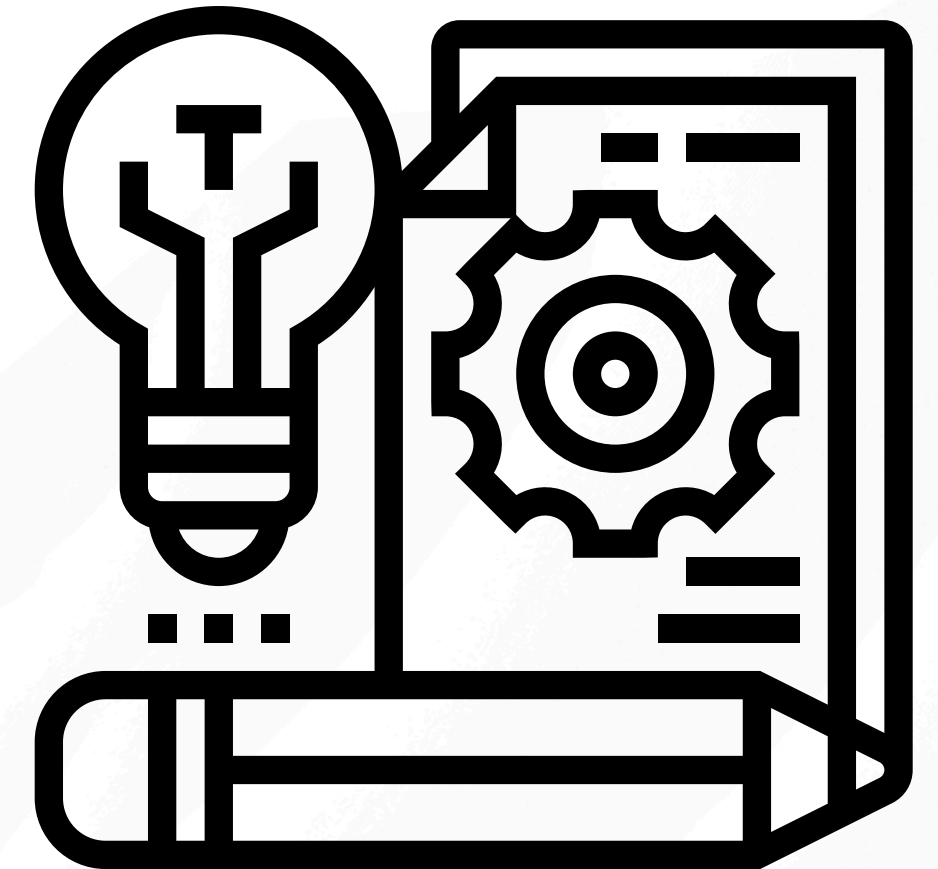


Adaptive Learning Systems

Study	Adaptation Approach
Reddig et al. (2025)	Individualised feedback based on student errors
Villegas-Ch et al. (2025)	Personalised adaptive learning
Lee et al. (2026)	Just-in-time adaptive feedback



Existing approach:

Individual learner → analyse individual performance → personalised feedback

Proposed approach:

Whole classroom → aggregate “Understand / Not Understand” → identify confusing slide → alert lecturer

- To support the lecturer during a live lecture, rather than provide personalised tutoring to each student.
- There is also supporting research showing that real-time class-level information can help teachers make instructional decisions
- For example, classroom communication systems have been used to collect students' answers and feedback the class results to trigger further discussion or teaching action.